



VINASAI INC

CONFIDENTIAL

SOFTWARE REQUIREMENTS SPECIFICATION

Examination Management System

Prepared for International Centre for Fine Arts

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1. Introduction & Project Overview

Purpose: This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the **Music Examination Management System** project.

1.1 Project Overview

This system is a comprehensive online platform for managing music examination registrations, application workflows, payment processing, examination scheduling, result publishing, and certification ceremonies. The platform accommodates multiple user types: including applicants, teachers, invigilators, evaluators, administrators, and super administrators—each with distinct permissions and access scopes. Applicants can register, manage multiple applications (for themselves or others), make bundled payments for multiple levels, and track progress through to result publication and ceremony scheduling. Teachers review and endorse applications, invigilators verify candidates at exam sessions, evaluators input and submit marks, and administrators oversee the entire exam lifecycle, including configuration, analytics, and communication workflows. The system enforces strict eligibility, scheduling, and grouping constraints to ensure fairness, transparency, and operational efficiency. All communications are handled exclusively via email.

1.3 User Classes and Characteristics

The following user roles have been identified for the system. Each role represents a distinct class of users with specific permissions and capabilities.

ROLE	DESCRIPTION	KEY PERMISSIONS
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Applicant	User who registers and manages applications for themselves or others, makes payments, views admission codes, receives email notifications, and views statuses and results.	• Register and verify account via email • Create, manage, and submit multiple applications • Select or invite teachers for each application • Make payments (Individual or bundled) and view payment status • View application statuses, admission cards/codes, results, and ceremony slot details
Teacher	User who reviews applications, can approve or reject them, and manages student lists and exam slots.	• View assigned applications in FIFO (First in First Out) order • Review candidate details and uploaded documents • Raise queries or correction requests to applicants • Reject or approve applications with esignature • View student list and assigned exam slots
Invigilator	User who verifies candidates at the exam hall using admission codes and marks attendance.	• Authenticate into invigilator dashboard • Search candidates by admission code (manual or scan) • View candidate verification profiles for assigned sessions • Mark admission codes as used/consumed (Single-use enforcement) • Restricted to candidates within scoped exam sessions
Evaluator (Examiner)	User who enters marks, assigns grades, and submits results for review.	• View candidates in assigned exam instances • Enter marks, assign grades, and record Pass/Fail

		• Save result entries as draft • Submit completed results to administrator • Cannot access data outside assigned instances
Administrator	User with full management capabilities over applications, results, and system configurations.	• Manage all applications and oversee workflows • Configure examination levels, fees, slots, venues, and buffer durations • Manage teacher and evaluator accounts • Review, correct, publish, and unpublish results • Manage ceremony session blocks and view/export attendance lists
Super Administrator	User with all administrator capabilities plus system level configuration and user management.	• All Administrator permissions • User and role management across all roles • System-level configuration and settings • Access and manage audit logs • Override and recover from system-level errors

2. Overall Description & Scope

2.1 Product Scope

The system will provide secure user registration and email verification, multicandidate application management, eligibility enforcement, teacher assignment and review, payment processing (including bundled payments), automated slot allocation with conflict prevention, generation of unique admission codes and cards, dedicated invigilator and evaluator dashboards, result entry and publishing workflows, ceremony slot allocation with family grouping (enforced as a non-negotiable hard constraint), and a robust, logged email notification system with delivery status tracking and automated retry handling.

2.2 Operating Environment

The system is designed as a web-based application accessible via modern browsers (such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari) on both desktop and mobile devices. The platform will support the latest two major versions of these browsers, ensuring compatibility and security. No native mobile or desktop application is planned at this stage. Deployment is expected on a secure, scalable cloud infrastructure (e.g., AWS, Azure, or GCP), though the specific provider is not stated. All user roles will interact with the system through role-specific dashboards within the web application.

3. System Features & Epics

This section describes the high-level system features (epics). Each feature includes associated user stories in the standard "**As a / I want / So that**" format.

3.1 Account Registration & Email Verification

Allows users to register, verify their email, and gain access to the system.

User Stories

ID	USER STORY
US - 1.1	As an Applicant , I want to register and verify my account via email , so that I can securely access the system and manage applications .
US - 1.2	As an Administrator , I want to view the status of user registrations and verifications , so that I can monitor account activation and intervene if needed .

3.2 Application Creation & Submission

Users can create and submit multiple competition applications (including registering multiple children or students under one account), including candidate and guardian information, and e-signature.

User Stories

ID	USER STORY
US - 2.1	As an Applicant I want to create and submit multiple applications for different candidates so that I can register myself, my children, or students for examinations .
US - 2.2	As an Applicant I want to submit required details and provide an e-signature so that my application is complete and verifiable .
US - 2.3	As a Teacher I want to be notified when an application is assigned to me so that I can begin the review process promptly .

3.3 Eligibility Enforcement Engine

Ensures candidates meet age and level prerequisites before application submission. The system supports a bundled registration workflow that allows applicants to register for all missing prerequisite levels in a single transaction.

User Stories

ID	USER STORY
US - 3.1	As an Applicant , I want to be prevented from submitting ineligible applications (e.g., too young or skipping levels) , so that I only submit valid applications .
US - 3.2	As an Applicant , I want to register for all missing prerequisite levels in a single bundled workflow , so that I can fulfill all eligibility requirements efficiently .
US - 3.3	As an Administrator , I want to configure eligibility rules and monitor enforcement , so that the system remains compliant with exam policies .

3.4 Payment Processing

Handles payment for examination levels, including bundled registrations.

User Stories

ID	USER STORY
US - 4.1	As an Applicant , I want to pay for all required examination levels in a single transaction , so that my applications are processed efficiently and I receive confirmation .
US - 4.2	As an Administrator , I want to configure fees and view payment statuses , so that I can track and reconcile payments .

3.5 Teacher Review Workflow

Allows teachers to review applications, query, reject, or approve them with an esignature.

User Stories

ID	USER STORY
US - 5.1	As a Teacher , I want to review applications in strict FIFO order and provide feedback or approval , so that I can ensure candidate eligibility and application completeness .
US - 5.2	As an Applicant , I want to receive notifications if my application is queried or rejected , so that I can respond or correct issues promptly .

3.6 Intelligent Slot Allocation Engine

Automatically assigns examination slots based on constraints like teacher cohort grouping and conflict prevention (candidate registered for two instruments/levels (e.g., Piano Level 2 and Violin Level 1) must never be assigned two slots that clash in time).

User Stories

ID	USER STORY
US - 6.1	As an Administrator , I want to automate slot allocation with teacher grouping and conflict prevention , so that the schedule is efficient and error-free .
US - 6.2	As an Applicant , I want to receive my assigned exam slot details promptly after approval , so that I can plan for the examination .

3.7 Admission Code & Card Generation

Generates unique admission codes and cards for candidates after slot allocation.

User Stories

ID	USER STORY
US - 7.1	As an Applicant I want to receive an admission card with a unique code and exam details so that I can gain entry to the exam session .
US - 7.2	As an Invigilator I want to verify candidates using admission codes so that only eligible candidates are allowed entry .

3.8 Examination Hall Verification

Allows invigilators to verify candidates using admission codes and mark attendance. Administrators can scope invigilators to specific exam instances to prevent cross session lookups.

User Stories

ID	USER STORY
US - 8.1	As an Invigilator , I want to search and verify candidates by admission code , so that I can mark attendance and prevent unauthorized entry .
US - 8.2	As an Administrator , I want to restrict invigilator access to assigned exam sessions , so that candidate data privacy is maintained .
US - 8.3	As an Administrator , I want to assign invigilators to specific exam instances , so that they cannot look up candidates from other sessions .

3.9 Result Entry & Publishing

Enables evaluators to enter results and administrators to review and publish them.

User Stories

ID	USER STORY
US - 9.1	As an Evaluator , I want to enter marks and submit results for my assigned candidates , so that results can be reviewed and published efficiently .
US - 9.2	As an Applicant , I want to view and download my official results after publication , so that I have proof of my performance .

3.10 Certification Ceremony Scheduling

Schedules passing candidates into ceremony slots based on account-level grouping. The family grouping for ceremony slots is enforced as a non-negotiable hard constraint.

User Stories

ID	USER STORY
US - 10.1	As an Administrator , I want to allocate ceremony slots to family groups of passing candidates, enforcing grouping as a hard constraint , so that families attend together and logistics are streamlined .
US - 10.2	As an Applicant , I want to be notified of my ceremony slot and details , so that I can attend the certification ceremony .

3.11 Email Notification System

Sends email notifications for various events in the application process. The system logs delivery status, supports automated retry handling for failed deliveries, and provides detailed specifications for the format and content of each type of email notification.

User Stories

ID	USER STORY
US - 11.1	As an Applicant , I want to receive timely email notifications at each important stage , so that I am informed and can take necessary actions .
US - 11.2	As an Administrator , I want to view email delivery logs and status tracking , so that I can ensure communications are reliably delivered .

4. Functional Requirements

ID	REQUIREMENT	ACCEPTANCE CRITERIA
FR-001	The system shall require all new users to verify their email address via a secure, time limited verification link before granting access to any features beyond the verification prompt.	• A verification link is sent immediately upon registration. • Users cannot access dashboard features until their email is verified.
FR-002	The system shall allow applicants to create and manage multiple independent applications, each representing a different candidate.	• An applicant can add, edit, or delete applications for different candidates under their account. • Each application is stored independently and can progress through workflows separately.
FR-003	The system shall enforce a minimum candidate age of 6 years at the time of application submission, rejecting ineligible applications with a clear message.	• Applications for candidates younger than 6 years are blocked from submission. • A user-facing error message explains the reason for rejection.
FR-004	The system shall enforce sequential level registration, blocking applications that skip prerequisite levels unless all missing levels are co-registered in a bundle.	• Applicants cannot select a higher level unless all lower levels are already passed or included in the current registration bundle. • The system offers bundled registration for all missing prerequisite levels.

FR-005	The system shall require applicants to select a registered teacher per application.	• Applicants cannot submit an application without selecting a teacher.
FR-006	The system shall collect and store an applicant's esignature during application submission.	• Applicants must provide an e-signature before submitting an application. • The e-signature is stored as part of the application record.
FR-007	The system shall process payments for examination levels, including bundled registrations, and mark applications as incomplete until payment confirmation.	• Applicants can pay for multiple levels in a single transaction. • Applications do not proceed to teacher review until payment is confirmed.
FR-008	The system shall allow applicants to retry payment upon failure, preserving all entered data without loss.	• A clear error message is displayed on payment failure. • All previously entered application data is retained for retry.
FR-009	The system shall queue applications for teacher review in strict FIFO (first in, first-out) order and prevent teachers from reordering or skipping entries.	• Teachers see applications only in submission order. • Teachers cannot reorder or skip applications in their review queue.

FR-010	The system shall allow teachers to raise queries or correction requests on applications, pausing the application and notifying the applicant.	• Teachers can send queries with comments to applicants. • Applicants receive notifications and the application is paused until a response is received.
FR-011	The system shall allow teachers to reject applications, requiring a written reason, which triggers a refund process and notifies the applicant.	• Teachers must provide a rejection reason. • The system initiates a refund process and notifies the applicant upon rejection.
FR-012	The system shall require teachers to provide an e-signature when approving an application, storing it with the application record.	• Teachers must provide an e-signature to approve applications. • The signature is stored and included in the admission card.
FR-013	The system shall automatically allocate examination slots to approved applications, grouping candidates by teacher and preventing overlapping slots for multi-instrument candidates.	• Candidates under the same teacher are grouped into the same exam instance as much as possible. • Multi-instrument candidates do not receive overlapping time slots, and a buffer is enforced between their slots
FR-014	The system shall generate a unique, non-guessable admission code for each candidate level combination after slot assignment.	• Each candidate-level receives a unique admission code with the specified format. • Admission codes are linked to the application, level, and assigned slot.

FR-015	The system shall generate a formatted admission card containing candidate, exam, and signature details, delivered via email and available for download.	• Admission cards include all specified fields and e-signatures. • Applicants receive the card by email and can download it from their dashboard.
FR-016	The system shall provide a dedicated invigilator dashboard for candidate verification by admission code, displaying full verification profiles and marking codes as used.	• Invigilators can search or scan admission codes and view candidate profiles. • Used or expired codes cannot be reused and display clear rejection messages.
FR-017	The system shall restrict invigilator access to candidates within their assigned exam sessions only.	• Invigilators can only view and verify candidates for sessions they are assigned to. • Attempts to access other sessions are blocked.
FR-018	The system shall allow evaluators to enter marks, assign grades, record Pass/Fail, save drafts, and submit results for administrator review for assigned exam instances.	• Evaluators can only access candidates in their assigned instances. • Results can be saved as draft and submitted for review.
FR-019	The system shall allow administrators to review, correct, publish, and unpublish results, making them visible to applicants only after publication.	• Administrators can edit and publish or unpublish results. • Applicants can only view results after publication.

FR-020	The system shall allow applicants to view and download official result PDFs only after results are published.	• Applicants are notified and can access result PDFs after publication. • Result PDFs include all specified fields and official signature block.
FR-021	The system shall allocate ceremony slots exclusively to passing candidates, grouping candidates registered under the same applicant account into the same ceremony slot as a nonnegotiable hard constraint.	• Only passing candidates are assigned ceremony slots. • Candidates from the same applicant account are always grouped in the same ceremony slot, and this grouping cannot be overridden.
FR-022	The system shall notify applicants of their ceremony slot allocation via email and display ceremony details on their dashboard.	• Applicants receive ceremony slot details by email. • Ceremony details are accessible from the applicant dashboard.
FR-023	The system shall send email notifications for all key events in the application and examination process, logging delivery status and supporting automated retry.	• Delivery status is logged per application and failed deliveries are automatically retried. • Emails are sent for all defined trigger events with correct content.
FR-024	The system shall support automated retry handling for failed email deliveries, attempting redelivery up to three times at increasing intervals before flagging as undeliverable.	• If an email delivery fails, the system automatically retries up to three times at configurable intervals. • If all retries fail, the event is logged and flagged for administrator review.

FR-025	The system shall allow administrators to assign invigilators to specific exam instances, restricting their access to only those sessions and preventing cross session candidate lookups.	• Administrators can assign invigilators to one or more exam instances. • Invigilators cannot search or view candidates outside their assigned sessions.
FR-026	The system shall implement a bundled registration workflow that automatically detects all missing prerequisite levels for a candidate and allows the applicant to register for all required levels in a single transaction.	• When a candidate selects a higher level, the system identifies all missing prerequisite levels. • Applicants can add all missing levels to a single bundled registration and complete payment in one transaction.
FR-027	The system shall provide detailed specifications for the format and content of each type of email notification, including subject line, sender name, branding, and required information per event.	• Each email notification type (e.g., registration, application status, payment confirmation, slot allocation, result publication, ceremony invitation) has a documented template with required fields. • All emails include professional branding, clear subject lines, sender name, and concise, actionable content.

5. Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and performance targets the system must meet. Each requirement includes a measurable metric and target value where applicable.

ID	CATEGORY	REQUIREMENT	METRIC	TARGET
NFR-001	Performance	The system shall respond to all user dashboard actions within 2 seconds under a load of up to 500 concurrent users.	response time	< 2 seconds
NFR-002	Security	The system shall enforce strong password policies, encrypt sensitive data at rest and in transit, and require email verification before access.	password complexity, encryption standards, verification status	NIST password guidelines, AES-256 at rest, TLS 1.2+ in transit, 100% verified users
NFR-003	Scalability	The system shall support scaling to accommodate up to 10,000 registered users and 2,000 concurrent application submissions	registered users, concurrent submissions	10,000 users, 2,000 concurrent submissions

		during peak periods.		
NFR-004	Usability	The system shall provide a user interface that enables first-time applicants to complete registration and application submission without external assistance in under 15 minutes.	first-time completion rate, time to complete	95% success, < 15 minutes
NFR-005	Reliability	The system shall achieve 99.5% uptime excluding scheduled maintenance windows.	uptime percentage	99.5% uptime

NFR-006	Maintainability	The system shall allow authorized administrators to configure examination levels, fees, time slots, and ceremony blocks without developer intervention.	admin configuration coverage	100% of configurable fields via admin UI
NFR-007	Compatibility	The system shall support the latest two major versions of Chrome, Firefox, Safari, and Edge browsers.	browser version support	Latest 2 major versions